WHIPSTOCK

RETRIEVABLE CASING SIDETRACK WHIPSTOCKS

Enedril

WHIPSTOCK

GENERAL DESCRIPTION

FAMILY OF RETRIEVABLE WHIPSTOCKS USED TO SIDETRACK OUT OF CASED HOLE, FOR RE-DRILLS, RE-ENTRIES AND MULTILATERAL WINDOWS. THE WHIPSTOCK IS RUN ON A STARTER MILL TOGETHER WITH ITS ANCHORING DEVICE, ORIENTED, SET, AND THE STARTER MILL IS SHEARED OFF THE WHIP; THE WINDOW IS THEN MILLED AND POLISHED ALONG THE WHIP FACE. THREE SETTING SYSTEMS ARE COVERED: A COMBINATION PACKER AND WHIPSTOCK (SET HYDRAULICALLY, SEALS THE HOLE), AN ANCHOR AND WHIPSTOCK (SET HYDRAULICALLY, NO SEAL) AND A MECHANICALLY SET BOTTOM-TRIP WHIPSTOCK. ALL HAVE A 3 DEGREE FACE ANGLE FOR POSITIVE KICK-OFF AND CAN BE RETRIEVED IN ONE TRIP. WINDOWS CAN BE MILLED THROUGH TWO STRINGS OF CASING.

TECHNICAL CHARACTERISTICS - PACKER TYPE

Casing size	Overall length	Body Ø	Packer length	Whip face length	Face angle (°)	Wt (lb)	Bypass valve length	Bypass valve Ø	Valve wt (lb)
5-1/2	182	4-3/16	76	106	3	585	25	3-3/8	40
7-5/8	216	5-5/8	84	133	3	980	43	4-3/4	150
7-5/8	229	5-15/16	84	146	3	1,400	43	4-3/4	150
9-5/8	261	8-5/8	84	178	3	2,500	46	6-3/4	240
13-3/8	338	11-3/4	87	251	3	6,595	38	8-1/4	400

TECHNICAL CHARACTERISTICS - MECHANICAL BOTTOM-TRIP TYPE

Casing size	Overall length	Body Ø	Anchor length	Whip face length	Face angle (°)	Wt (lb)
5 - 5-1/2	142-3/8	4-5/16	35-1/8	74	3	570
7	172-1/8	5-7/16	38-1/8	95	3	875
8-5/8	189-1/2	7-7/16	43-1/2	124	3	1,175

Dimensions in inches, weights in lb (approximate). Packer type: bypass valve data listed separately. Bottom-trip type can also be set permanently; 7-5/8 and 9-5/8 sizes on request.

DESIGN FEATURES

- **One trip:** anchor or packer, starter mill and whip run together; retrieved in one trip.
- **Anchor or packer:** large slip area reduces casing stress; ratchet ring and anti-rotation keys hold the whip.
- **Shear bolt:** confirms the anchor is fully set before milling starts.
- **Bottom-trip type:** sets with 3,600 lb on a trigger rod; no hydraulic pressure; slips retract on retrieval.
- **Orientation:** muleshoe sub, gyro or MWD.

SETTING PARAMETERS

Hydraulic set (packer, anchor)	3,000 - 3,500 psi
Bottom-trip set, shear-down	15,000 - 20,000 lb
Set depth	2 - 6 ft above a casing collar
Hole angle above 4°	Set within 90° of high side
Window milling	Starter, window, watermelon mills

HOW TO ORDER

Please specify: 1. casing size and weight 2. setting system (packer, anchor or bottom-trip) 3. connections 4. orientation method.